



Emerging diseases caused by Geminiviridae

A review paper written by Varma & Malati (2003) provides a recent update on the diseases which are caused by *Geminiviridae*. During the last two decades, these viruses have emerged as devastating pathogens, particularly in the tropics and subtropics, causing huge economic losses and threatening crop production. Among *Geminiviridae*, begomoviruses are causing the most serious problems in a variety of crops. The emergence of new begomoviruses and their spread has been favoured by the appearance of *Bemisia tabaci* biotype B, the increase of its populations, and the evolution of virus variants. Variability arose though mutations, recombination and pseudorecombination, not only among variants but also among species and even genus. Human activities have also played an important role in disseminating viruses (movements of infected planting material) or creating more favourable conditions for them (e.g. changes in cropping systems, introduction of new crops). The most important diseases caused by *Geminiviridae* are reviewed.

- Mastreviruses: Maize streak virus, Bean yellow dwarf virus, Chickpea chlorotic dwarf virus
- Curtoviruses: Beet curly top virus
- Begomoviruses: Cassava viruses, Cotton leaf curl viruses, Cucurbit viruses, Legume viruses (bean golden mosaic disease, cowpea golden mosaic disease, yellow mosaic disease), begomoviruses of potato, sweet potato, tomato (with a useful list of virus species found on this crop).

Sources

Varma, A.; Malathi, V.G. (2003) Emerging geminivirus problems: a serious threat to crop production. **Annals of applied Biology**, 142(2), 145-164.



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